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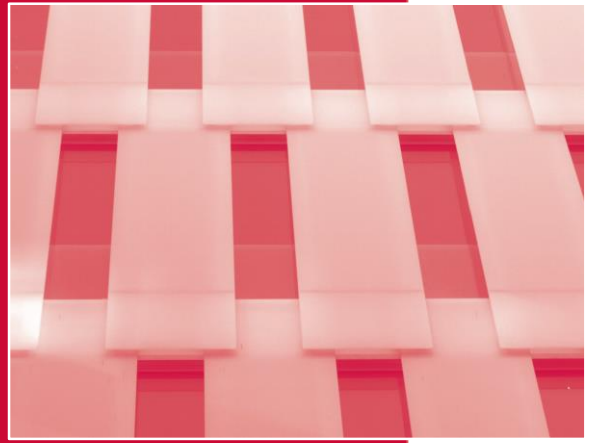
Financial Accounting – Course 30693



Financial Statement presentation, reclassification and analysis

Part 3

Università Bocconi
UNDERGRADUATE SCHOOL

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AGENDA

1. Understanding the Business
2. External decision-makers. F/S presentation
3. Internal decision-makers
4. F/S reclassification. Key indicators
5. Ratio Analysis
6. FSA in the real-life world
   - 6A. Forecast, budget and business plan
   - 6B. Rating agencies and analyst reports
   - 6C. The valuation multiples
   - 6D. Focus on EBITDA, EBIT & NFP

PART 3

6A. Forecast, budget and business plan

Budgeting and business plans

Mastery of Accounting is necessary to prepare three fundamental documents:

(1) Forecast consists of predicting the income statement and cash flow statement for the current year (e.g., in March 2024, the forecast includes monthly or quarterly projections through December). The balance sheet is sometimes excluded, as *the purpose is to monitor short-term performance, particularly (i) monthly net income and (ii) available cash.*

(2) Budget represents the forecast of the Income Statement (IS), Balance Sheet (BS), and Cash Flow Statement (CFS) for the subsequent accounting period. It is usually prepared in the second half of the year (e.g., in October 2024, the 2025 budget is approved by the Board). *Its purpose is to set financial targets for the following year.*

(3) Business Plan is a key document that provides forecasts of IS, BS, and CFS over the next three to five years. It serves multiple purposes, including assessing the soundness of the company's strategy, supporting impairment testing through the estimation of future cash flows, guiding long-term investment decisions (e.g., evaluating whether a new production plant will generate profits and cash inflows), informing debt restructuring plans, and supporting extraordinary transactions. It is also used to estimate the company's market value. Overall, *its primary objective is to project how the financial statements will evolve in the coming years.*

All three documents are prepared following a careful analysis of historical IS, BS, and CFS data, as well as component percentage calculations. These documents, especially the Business Plan, are formally approved by the Board of Directors.

An example of a Business Plan prepared in 2010 (Budget 2011 plan from 2012 on)

€/'000	Bdg FY11	Plan FY12	Plan FY13	Plan FY14	Plan FY15
Branded sales	167.729,7	191.466,8	212.621,9	234.562,4	240.375,6
Other sales	1.710,7	1.914,7	2.126,2	2.345,6	2.403,8
Total Sales	**169.440,4**	**193.381,4**	**214.748,1**	**236.908,0**	**242.779,4**
Chassis CoGS	(74.472,6)	(84.802,1)	(94.235,7)	(103.980,0)	(106.519,0)
Other mat costs	(56.387,5)	(64.435,2)	(71.351,5)	(78.560,9)	(80.528,2)
CoGS	**(130.860,0)**	**(149.237,3)**	**(165.587,1)**	**(182.540,9)**	**(187.047,2)**
Other costs	(2.010,5)	(2.010,3)	(2.232,4)	(2.462,8)	(2.523,8)
Total CoGS	**(132.870,5)**	**(151.247,6)**	**(167.819,6)**	**(185.003,7)**	**(189.571,0)**
Variable Margin	**36.569,9**	**42.133,8**	**46.928,5**	**51.904,3**	**53.208,4**
Sales & Mkt Expenses	(3.561,1)	(3.869,0)	(4.069,0)	(4.269,0)	(4.269,0)
Other indirect expenses	(5.503,4)	(4.883,4)	(4.883,4)	(4.883,4)	(4.883,4)
Semi-variable costs	**(9.064,5)**	**(8.752,4)**	**(8.952,4)**	**(9.152,4)**	**(9.152,4)**
Direct Plant Labor	(10.744,6)	(11.764,8)	(12.570,7)	(13.844,8)	(14.142,7)
Indirect Plant Labor	(2.985,9)	(2.959,5)	(3.054,0)	(3.189,2)	(3.249,8)
Personnell Commercial	(2.437,5)	(2.368,9)	(2.276,1)	(2.247,5)	(2.247,5)
HQ - Personnell Commercial	(101,2)	(75,5)	(75,5)	(75,5)	(75,5)
Personnell R & D	(1.133,8)	(1.118,3)	(1.118,3)	(1.118,3)	(1.118,3)
Personnell G & A	(2.443,8)	(2.297,4)	(2.070,5)	(1.973,8)	(1.973,8)
Personnell costs	**(19.846,8)**	**(20.584,4)**	**(21.165,1)**	**(22.449,1)**	**(22.807,7)**
Other fixed costs	(4.535,1)	(4.856,1)	(4.656,1)	(4.656,1)	(4.656,1)
EBITDA	**3.123,5**	**7.940,9**	**12.154,9**	**15.646,7**	**16.592,3**
EBITDA %	*1,8%*	*4,1%*	*5,7%*	*6,6%*	*6,8%*
Plant Depreciation	(2.672,9)	(2.848,3)	(3.023,7)	(3.199,1)	(3.374,5)
Other depreciation	(1.672,1)	(2.182,1)	(2.292,1)	(2.402,1)	(2.512,1)
EBIT	**(1.221,5)**	**2.910,5**	**6.839,1**	**10.045,5**	**10.705,6**
EBIT %	*(0,7%)*	*1,5%*	*3,2%*	*4,2%*	*4,4%*
Financial items	(2.064,4)	(2.979,7)	(3.109,7)	(3.146,3)	(3.018,1)
Earnings before Extr. Costs	**(3.286,0)**	**(69,2)**	**3.729,4**	**6.899,2**	**7.687,5**
Extr. Expenses / Incomes	(1.330,4)	-	-	-	-
Goodwill Amortization	(1.279,3)	(949,3)	(949,3)	(949,3)	(949,3)
IBT	**(5.895,7)**	**(1.018,5)**	**2.780,0**	**5.949,9**	**6.738,2**
Taxes	-	(1.684,4)	(1.113,6)	(1.682,3)	(1.861,2)
Net Income	**(5.895,7)**	**(2.702,9)**	**1.666,5**	**4.267,6**	**4.877,0**

6B. Rating agencies and analyst reports

Rating agencies and analyst reports

Rating agencies and Financial Analysts are:

- Expert in firms' valuation
- Highly visible and qualified
- Provide Recommendations and Target Prices

What do they do?

- They "Filter" the information released by the firms and insert it in their own evaluation process.
- The elaboration of the information end with the output of the specific activity of an analyst: the production of the analyst reports.

Analyst reports (also called Equity research report) is an analyst-authored document that evaluates a publicly traded company and advises investors on whether to purchase, retain, or sell its shares.

Buy-side firms

- Such as asset management companies, maintain in-house research teams that generate analyses and recommendations on which securities portfolio managers should purchase or divest. These reports are intended strictly for internal use to support investment decisions and are not shared with the public.

Sell-side firms

- Like investment banks produce equity research reports for external distribution to clients in sales and trading as well as wealth management. These reports are typically provided at no cost and include explicit ratings (buy, sell, or hold) along with a projected target price.

Analyst reports (Forecast Firms' Future Value)

The analyst's work can be represented as a "*step by step*" procedure:

- **from the *collection* and the *elaboration* of the information,**
- **to the *conversion of the forecasts* into the estimation of the *firm value***
- **and the formulation of *target prices* and *stocks recommendations* that can influence the market behavior.**

The main output is a simple question with a multifaceted – yet always incomplete – answer:

Would you invest in Company X if you had funds available to do so?

Three main types of analyst report:

Initial Coverage

first time an analyst issues a report

Analyst Report

regular reports on a constant basis

Updates

needed when firms' related event might influence firms' value.

6C. Valuation Multiples

Valuation Multiples

Valuation multiples are one of the two most common methods used to estimate the market value of a company (i.e., to value a company), starting from the data presented in the financial statements.

Valuation multiples are a method of obtaining an initial, broad, and approximate estimate of a company's value, as follows:

First, you select an accounting item from the company's financial statements to be evaluated (the "**Target**") *(e.g., net income, also referred to as net earnings).*

Second, you select a sample of comparable companies operating in the same industry, with publicly available data *(e.g., listed companies)*, and compute a ratio in which the numerator is the market value and the denominator is the same accounting item selected for the target company *(e.g., the Price/Earnings, or P/E, ratio).*

Third, you multiply the accounting item selected for the target by the corresponding market multiple; this yields the estimated market value of the target company.

Valuation Multiples (Example 1/2)

ABC is a company operating in the consumer healthcare products sector.
ABC's average earnings over the past three years, as well as the projected earnings for the next three years according to the business plan, are €70,000 with no significant variation.

What is the market value of ABC's shares if a potential investor intends to acquire the company?

Assume that Pfizer's P/E ratio is in line with that of its competitors and is reliable for this purpose.

Pfizer PE Ratio 2010-2023 | PFE

Prices | Financials | Revenue & Profit | Assets & Liabilities | Margins | Price Ratios | Other Ratios | Other Metrics

PE Ratio | P/S Ratio | Price/Book Ratio | Price-FCF Ratio | Net Worth

Current and historical p/e ratio for Pfizer (PFE) from 2010 to 2023. The price to earnings ratio is calculated by taking the latest closing price and dividing it by the most recent earnings per share (EPS) number. The PE ratio is a simple way to assess whether a stock is over or under valued and is the most widely used valuation measure. Pfizer PE ratio as of March 15, 2024 is **15.27**.

Please refer to the Stock Price Adjustment Guide for more information on our historical prices.

Valuation Multiples (Example 2/2)

In light of the data provided above, the market value of ABC's shares is:

€ 70,000 x 15.27
= € 1,068,900

Speaking in general terms, and purely for illustrative purposes, using only the data provided in this example, a buyer could offer the shareholders a price between €1 million and €1.1 million for the acquisition of 100% of the company's shares.

Valuation Multiples

This is an Accounting course, not a finance and/or valuation course. The objective is simply to highlight that, when the market value of a company is derived from its financial statements, those statements must be reliable and properly adjusted ("normalized"). Only with a solid understanding of Accounting can such an evaluation be considered reliable.

The following are a few examples illustrating this point:

- What are the company's depreciation policies?
- Has any impairment loss been recorded?
- Are there any non-recurring, unusual, or exceptional revenues or expenses?
- Have any recurring but non-monetary items been recorded (e.g., allowance for doubtful accounts)?
- Have all adjusting entries been properly recorded?
- Are the targets in the business plan reasonable or overly ambitious?
- Are revenues properly recognized?
- Are all estimates (e.g., net receivables, inventory) properly measured?

Valuation of companies in an M&A transaction

As stated previously, this is not a valuation course; however, it is useful to highlight the two most important figures in financial statements that are commonly used worldwide to estimate (broadly) the market value of companies.

Before addressing this, it is important to note that the market value of a company can be estimated either on an Asset Side basis or on an Equity Side basis.

Asset side

- when you calculate the market value of a company asset side you are calculating the **ENTERPRISE VALUE**, which is the market value of the assets of a company i.e., the amount that a buyer should pay to purchase all the assets of a company together, leaving then the target as a "box" full only of cash and debt

Equity side

- when you calculate the market value of a company equity side you are calculating the **EQUITY VALUE**, which is the market value of the shares of a company.

The Enterprise Value (EV) or the Equity Value (EqV) can be calculated using two methods (among the others but with the highest frequency):

- ✓ Discounted Cash Flow (DCF): first you prepare a Business Plan, then you calculate free cash flows and then you calculate the present value of the cash flows in the plan, plus a terminal value
- ✓ Multiples: see preceding slides

Valuation of companies in an M&A transaction

To calculate the Enterprise Value and the Equity Value, the two most important data of the financial statements are:

EBITDA and NET FINANCIAL POSITION (NFP)

Equity Value = Enterprise Value – Net Financial Position

Someone who knows the importance of EBITDA …

Source: LinkedIn

The most frequent multiples

Following are some of the most frequent valuation multiples.

ASSET SIDE

- ✓ EV/Sales: enterprise value is calculated as a multiple of the sales. Example: pharmacies, between 1.3x and 1.5x
- ✓ EV/EBITDA: this is the most common valuation multiple used worldwide; enterprise value is calculated as a multiple of EBITDA. Example: healthcare sector between 10x and 15x
- ✓ EV/EBIT: enterprise value is calculated as a multiple of EBIT.

(remember that if you use an asset-side multiple, to calculate the value of the shares you have to subtract Net Financial Position or add it, if cash > financial debt).

EQUITY SIDE

- ✓ P/E and P/BV: referred to as market ratios (see next slide)

Market Ratios

Price-to-Earnings Ratio (P/E)

$$\frac{\text{Market Price per share}}{\text{EPS}}$$

- Measures the relationship between current market price and EPS (Earnings Per Share)
- Market price reflects the expectations of the market for future earnings
- It is based on future subjective expectations, not on actual past figures

Price-to-Book Value Ratio (P/BV)

$$\frac{\text{Market Price per Share}}{\text{Book Value per Share}}$$

- Measures the relationship between the current market price and the book value per share (= book value of total shareholders'equity)
- Shows how much investors are willing to pay for each unit of stockholders'equity
- Reflects market expectations about the company's asset quality, profitability, and future growth
- Based on accounting net worth, so it is more balance-sheet focused than earnings focused
- Useful especially for asset-heavy businesses such as banks, financial institutions, and manufacturing firms

6D. Focus on EBITDA, EBIT & NFP

Focus on EBITDA (and EBIT)

In the first FSA session in the standard format example of an Income Statement format, we calculated:

EBITDA AND EBIT → They are both measures of a business's profitability

Because EBIT and EBITDA are independent of discretionary capital structure decisions (i.e., the financing mix) and are pre-tax measures of profitability, they can be used for comparability purposes across different companies. In particular, EBITDA is a metric used to evaluate a company's operating performance, indicating whether the company is able to generate operating income and cash flows, excluding investment, extraordinary, financing, and tax effects.

The magnitude of the difference between EBIT and EBITDA depends on the industry in which the company operates, as well as on discretionary management decisions (e.g., the choices of capitalizations and/or the assumed useful life of fixed assets (PP&E) and management's efficiency in capital expenditures, or lack thereof).

The difference between EBIT and EBITDA has further increased in recent years due to the growing use of the "Adjusted EBITDA" metric, which extends traditional EBITDA through additional discretionary adjustments.

The rationale for these adjustments, at least in theory, is to present a company's operating performance more accurately and provide investors with greater transparency, albeit at the cost of increased managerial discretion and reduced standardization.

Focus on EBITDA (and EBIT)

In practice, EBIT and EBITDA are two of the most widely used metrics, particularly for performing valuation analyses, such as in the context of fundamental investing in public markets and mergers and acquisitions (M&A) transactions.

The use of EBITDA is more common in M&A transactions because it is broadly aligned with the target's cash flow profile* and ultimately translates into operating cash flow, which is the key variable being negotiated (subject to mutually agreed adjustments).

**Please note that the target's cash flow profile must be normalized in order to produce reliable cash flow projections; all distorting elements to be included or excluded, thereby leading to normalized EBITDA, should be clearly identified.*

EBITDA vs. Cash (A Simple Example)

Assume a consulting firm performs services worth 100 each year and incurs cash operating expenses of 40 each year. In Year 1, the firm invoices the customer for 100 but only collects 80 in cash before year-end, so 20 remains in accounts receivable. In Year 2, the firm again earns revenue of 100, however, collects it all and it also collects the 20 receivable from Year 1.

	Year 1	Year 2	Total
Revenue	100	100	200
Operating expenses	(40)	(40)	(80)
EBITDA	60	60	120
Accounts receivable	20	(20)	0
Cash collections	80	120	200
Cash payments	(40)	(40)	(80)
Operating cash flow	40	80	120

This shows why EBITDA can be viewed as a proxy for cash flow over time: working capital movements can create short-term timing differences, but once those timing effects reverse, the difference disappears.

Focus on EBITDA (and EBIT)

EBIT and EBITDA metrics are frequently used to perform relative valuation, in which the two metrics coincide with the enterprise value (TEV) metric.

$$\frac{\text{EV (Enterprise value)}}{\text{EBITDA}}$$

$$\frac{\text{EV (Enterprise value)}}{\text{EBIT}}$$

Two of the most common valuation multiples used in relative valuation – such as comparable companies' analysis (or "trading comps") and precedent transactions analysis (or "transaction comps")

For a valuation multiple to be practical, the numerator (i.e. the value measure) and denominator (i.e. the value driver) must match concerning the stakeholders represented.

Hence, EBIT and EBITDA both correspond to enterprise value, rather than equity value, since the profit metrics have not yet been adjusted for any payments to lenders, namely the periodic interest expense payment owed on outstanding debt.

Focus on NFP

As stated previously, NFP is subtracted from Enterprise Value to compute Equity Value. It is calculated as follows:

NFP = financial liabilities - cash and cash equivalent

It is measured at the end of each reporting period (typically annually or quarterly). Financial assets and liabilities are valued at market value.

Nonetheless, within an M&A transaction, several elements affect this value; in this context, the indicator is referred to as normalized NFP.

Please note: In M&A transactions, Financial Due Diligence involves an in-depth examination and analysis of, among other aspects, three key elements: EBITDA, Net Financial Position, and Net Working Capital. These elements constitute the basis for determining the value of a company (or group of companies or business unit) that is the subject of the transaction.

Final remarks

Once again, and in conclusion, we emphasize that if the market value of a company is calculated based on accounting figures, a solid understanding of Accounting is essential to make sound investment, strategic, operating, M&A, and financing decisions.

Indeed, accounting policies (or accounting fraud) can have a significant impact on the returns of all stakeholders involved. Consider the following examples:

What if the inventory's book value must be written down?

What if too many expenses have been capitalized, thereby affecting EBITDA?

What if an impairment test is overly optimistic and an intangible asset is not impaired?

What if a debt instrument is not properly measured?

What if an unusual transaction is not disclosed as such in the notes, and is therefore included in normalized EBITDA?

What about the calculation of income taxes and contingent liabilities?

What if net accounts receivable are not properly measured (i.e., errors in bad debt expense)?

What if VAT was non-deductible?

Knowledge of the correct accounting rules for recording and measurement will guide all of you, as future CEOs, CFOs, consultants, analysts, fund managers, and advisors, in making sound decisions.

Thank you!

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End of FSA slides